

Relation Algebra

§11. Relator

(11a)

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from R to S IS $R \sqsubseteq S$. A **relator** $F : \mathcal{C} \rightarrow \mathcal{D}$ is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving $\circ$ is **not** asked for — $\circ$ is an identity-on-objects involution $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$, no part of the 2-category.

(11b)

the statement

$\text{Map}(f) \implies \text{Map}(F(f))$ and $\text{Map}(f) \implies F(f^\circ) = F(f)^\circ$.

Over a **tabular** allegory a functor is a relator $\iff$ it preserves $\circ$.

$F(R^\circ) = F(R)^\circ$ for every R , so $F(R)^\circ$ needs no bracket.

Two relators agreeing on maps are equal.

$\text{Coreflexive}(X) \wedge \text{Coreflexive}(Y) \implies F(X \cap Y) = F(X) \cap F(Y)$

$F(R \cap S) \sqsubseteq F(R) \cap F(S)$, and strictly.

$\text{PreservesRecip}(F) \implies F(\text{dom}(R)) = \text{dom}(F(R))$

The **power relator** $P \multimap x \sqsubseteq P(R) \ y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$ — is where the fourth is strict: for $R = \{(a_1, b_1), (a_2, b_2)\}$ and $S = \{(a_1, b_2), (a_2, b_1)\}$ the pair $(\{a_1, a_2\}, \{b_1, b_2\})$ is in $P(R) \cap P(S)$, while $R \cap S = \emptyset$.

§11.1. Fork $\langle R, S \rangle$

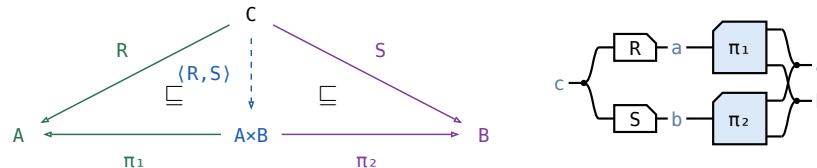
(11.1a)

The **fork** of $R : \mathcal{C} \rightarrow \mathcal{A}$ and $S : \mathcal{C} \rightarrow \mathcal{B}$ is $\langle R, S \rangle \triangleq R \pi_1^\circ \cap S \pi_2^\circ$, where (π_1, π_2) is the tabulation of τ .

(11.1b)

$$\langle R, S \rangle \pi_1 = \text{dom}(S) R \quad \langle R, S \rangle \pi_2 = \text{dom}(R) S$$

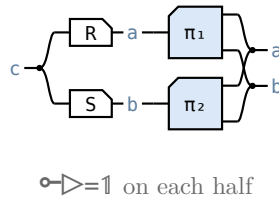
(11.1c)



A domain is coreflexive, so $\langle R, S \rangle \pi_1 \sqsubseteq R$, with equality exactly when S is entire; for maps both triangles commute and $\langle f, g \rangle$ is unique. In Rel , $c \sqsubseteq \langle R, S \rangle (a, b)$ iff $c R a$ and $c S b$ — copy c , then R on one strand and S on the other, which is $\triangleleft (R \otimes S)$ on the right.

No $\circ$ survives the translation. $\pi_1 = 1 \otimes \circ$ discards the second component, so $\pi_1^\circ = 1 \otimes \circ$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(11.1d)

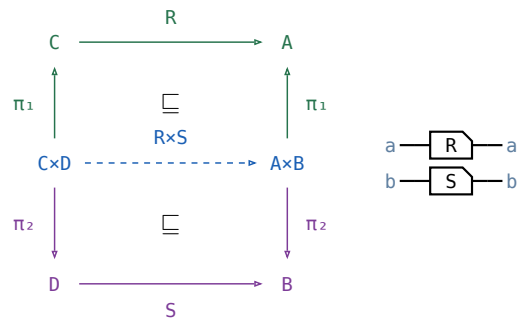


§11.1.1. Relational product $R \times S$

(11.1.1a)

$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$, a relator in each argument but no longer a categorical product.

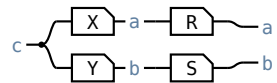
(11.1.1b)



Right-then-up is $(R \times S)\pi_1$, up-then-right is $\pi_1 R$, and $(R \times S)\pi_1 \sqsubseteq \pi_1 R$, equality when S is entire. In $\mathbf{Rel}$, $(c, d) (R \times S) (a, b)$ iff $c R a$ and $d S b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§11.1.2. Absorption

For $X : E \rightarrow C$ and $Y : E \rightarrow D$, $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$. Both sides are this picture:



(11.1.2a)

(11.1.2b)

the law	picture
1 $R \times S \triangleq (\pi_1 R, \pi_2 S)$	
2 $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ both sides are the same strokes — absorption. Its $\sqsubseteq$ half is that half at $S := \mathbb{1}$ and at $R := \mathbb{1}$ — stages of their proof of this row; the corollary is the $(R \times S) (U \times V) = (RU) \times (SV)$ it yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$.	
3 $\langle R, S \rangle \pi_1 = \text{dom}(S) R$ (11.1b)	
4 $\langle R, S \rangle \pi_2 = \text{dom}(R) S$ (11.1b)	
5 $\langle X, Y \rangle (R, S)^\circ = XR^\circ \cap YS^\circ$	
6 $\langle R, S \rangle^\circ (U, V) \sqsubseteq (R^\circ U) \times (S^\circ V)$	
7 $\text{Map}(f) \implies f \langle R, S \rangle = \langle fR, fS \rangle$ f a map; it fails for an arbitrary arrow;	
8 $F(R \times S) \text{unzip}(F) \sqsubseteq \text{unzip}(F) (F(R) \times F(S))$ $\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$; only $\sqsubseteq$ for an arbitrary R, S	
9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$ reading $\times$ as the relational product, does $\mathbf{Rel}$ have exponentials?	

§11.2. Coproduct $[R, S] : A + B \rightarrow C$

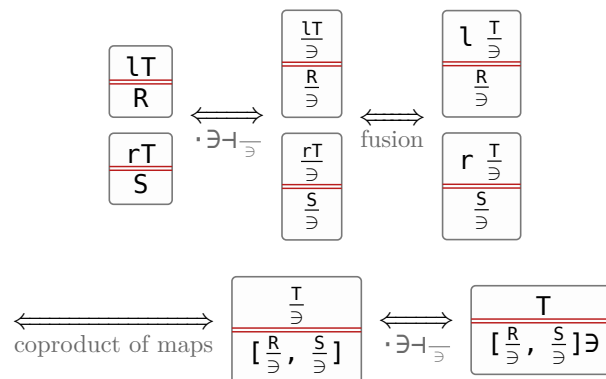
(11.2a)

the statement	picture
$[R, S] \triangleq l^\circ R \cup r^\circ S$ The tape is the union — a particle entering at A+B takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$	
$R + S \triangleq [Rl, Sr]$	
$l[R, S] = R, r[R, S] = S$ $lT = R \wedge rT = S \implies T = [R, S]$	
$ll^\circ = 1, rr^\circ = 1$	
$lr^\circ = \emptyset, rl^\circ = \emptyset$	
$l^\circ l \cup r^\circ r = 1$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$	

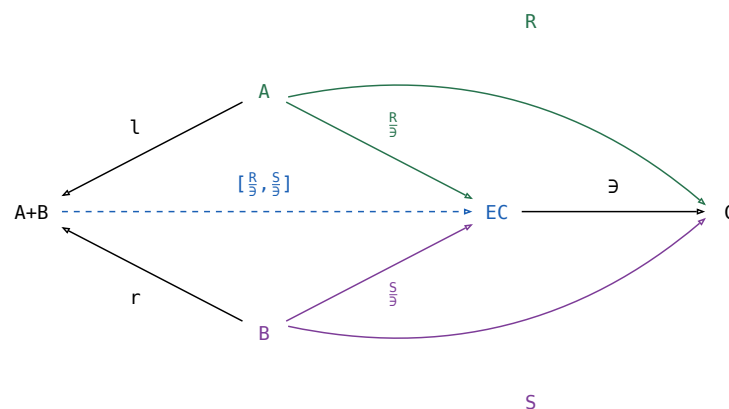
§11.2.1. $[R, S] \triangleq [\frac{R}{\exists}, \frac{S}{\exists}] \exists$

The universal-property row is not free: l, r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because $\frac{\exists}{\exists}$ sends an arrow $A \rightarrow C$ to a map $A \rightarrow EC$ reversibly, so the map coproduct can be applied underneath it. For any $T : A+B \rightarrow C$,

(11.2.1a)



(11.2.1b)



Nothing here holds only up to Ξ : every triangle commutes on the nose, which is the difference from the fork above. The border spells $[R, S] = [\frac{R}{\exists}, \frac{S}{\exists}] \exists$, and pushing $\exists$ into the union that $[\cdot, \cdot]$ on maps already is turns that back into the definition, $(\frac{R}{\exists} \cup \frac{S}{\exists}) \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists = \frac{R}{\exists} \exists \cup \frac{S}{\exists} \exists$.

(11.2.1c)

the law	picture
$[R, S] \triangleq \frac{R}{\exists} \cup \frac{S}{\exists}$ (11.2a)'s first row	
$R + S \triangleq [Rl, Sr]$	
$[U, V]^\circ [R, S] = U^\circ R \cup V^\circ S$ (11.2a)'s last row;	
$[1, 0] = \frac{1}{\exists}, [0, 1] = \frac{0}{\exists}$ $[\frac{1}{\exists}, \frac{0}{\exists}] = 1$ which is (5.9)	
prove (5.11), and say why duality does not carry it over from the product law	
$(R + S) \cap [P, Q] [U, V]^\circ =$ $(R \cap P U^\circ) + (S \cap Q V^\circ)$ $[P, Q] [U, V]^\circ$ is a full 2×2 of composites; $R + S$ is diagonal, so the meet cuts the two off-diagonal branches	

§11.3. The power relator $P(R)$

(11.3a)

For $R : A \rightarrow B$,

$$P(R) \triangleq (\epsilon \backslash R \epsilon) \cap (\exists R / \exists)$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R) / \exists) \cap (\exists / (\exists R))^{\circ}$$

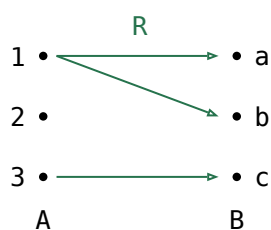
$$x \in P(R) \ y \iff (\forall a \in x. \exists b \in y. a R b) \wedge (\forall b \in y. \exists a \in x. a R b)$$

Every element of x is related by R to some element of y , and conversely.

(11.3b)

	in Rel	in words	$\text{im}(xs)$	partner
$(\exists R) / \exists$	$\forall y \in ys. \exists x \in xs. x R y$	$\forall y. \text{ some } xs R y$	$ys \subseteq \text{im}(xs)$	every y has a partner
$(\exists / (\exists R))^{\circ}$	$\forall y. (\exists x \in xs. x R y) \rightarrow y \in ys$	every y with some $xs R y$ is in ys	$ys \supseteq \text{im}(xs)$	ys leaves out no partner
$E(R)$	$ys = \{y \mid \exists x \in xs. x R y\}$	$ys = \text{every } y \text{ with some } xs R y$	$ys = \text{im}(xs)$	every y has a partner, and none is left out
$((\exists R^{\circ}) / \exists)^{\circ}$	$\forall x \in xs. \exists y \in ys. x R y$	$\forall x. x R \text{ some } ys$	—	every x has a partner
$P(R)$	$\forall y \in ys. \exists x \in xs. x R y$ and $\forall x \in xs. \exists y \in ys. x R y$	$\forall y. \text{ some } xs R y$ and $\forall x. x R \text{ some } ys$	—	every x and every y has a partner

(11.3c)



xs	$(\exists R) / \exists$	$(\exists / (\exists R))^{\circ}$	$E(R)$	$((\exists R^{\circ}) / \exists)^{\circ}$	$P(R)$
$\emptyset$	$\emptyset$	all 8 subsets of abc	$\emptyset$	all 8 subsets of abc	$\emptyset$
$\{1\}$	$\emptyset, a, b, ab$	ab, abc	ab	a, b, ab, ac, bc, abc	a, b, ab
$\{2\}$	$\emptyset$	all 8 subsets of abc	$\emptyset$	none	none
$\{3\}$	$\emptyset, c$	c, ac, bc, abc	c	c, ac, bc, abc	c
$\{1, 2\}$	$\emptyset, a, b, ab$	ab, abc	ab	none	none
$\{1, 3\}$	all 8 subsets of abc	abc	abc	ac, bc, abc	ac, bc, abc
$\{2, 3\}$	$\emptyset, c$	c, ac, bc, abc	c	none	none
$\{1, 2, 3\}$	all 8 subsets of abc	abc	abc	none	none

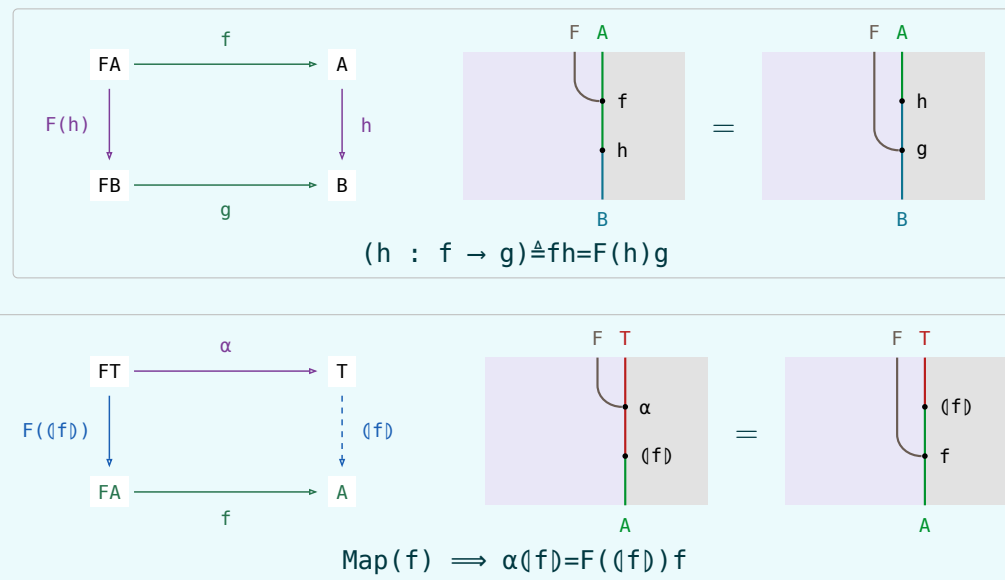
(11.3d)

law	the reading
$X \subseteq P(R) \iff X \subseteq \exists R \text{ and } X^{\circ} \subseteq \exists R^{\circ}$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^{\circ}) = P(R)^{\circ}$, and $R \subseteq S \implies P(R) \subseteq P(S)$.
$P(1) = 1$	The straightness axiom verbatim: extensionality is P 's unit law.
$\text{Map}(f) \implies P(f) = E(f)$	In Rel, $x \in P(f) \ y \iff ys = \{f(a) \mid a \in x\}$. The half at f° says every $a \in x$ has its $f(a)$ on ys ; f has just the one image per a , so that already says ys contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(RS) = P(R)P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$.

§11.4. Initial algebra

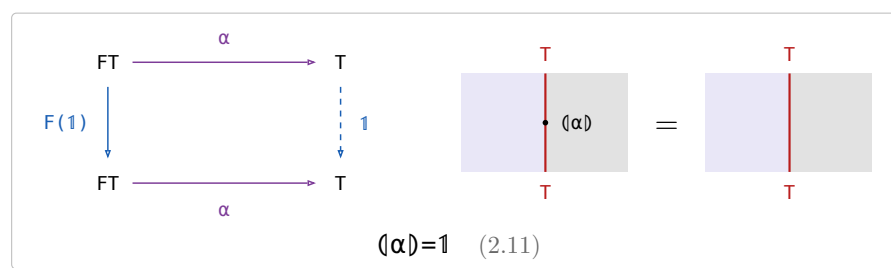
(11.4a)

An **F-algebra** is a map $f : F(A) \rightarrow A$; A is its **carrier**. An **F-homomorphism** from $f : F(A) \rightarrow A$ to $g : F(B) \rightarrow B$ is a map $h : A \rightarrow B$ with $f \circ h = F(h) \circ g$. The **initial algebra** $\alpha : F(T) \rightarrow T$ is the F-algebra with exactly one F-homomorphism $\langle f \rangle : T \rightarrow A$ to every F-algebra f .



§11.4.1. Reflection

(11.4.1a)

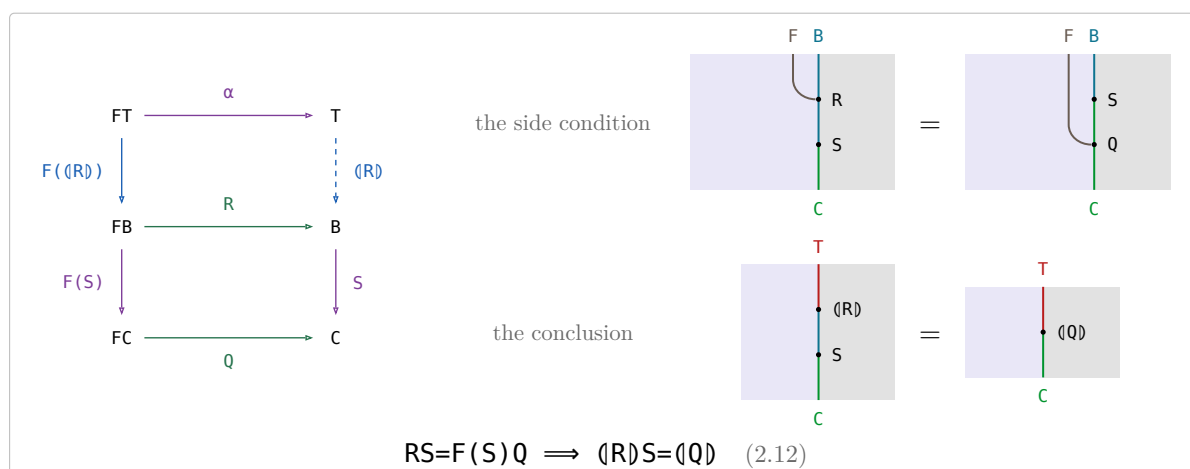


Taking a value apart with α and putting it straight back is doing nothing.

§11.4.2. Fusion: $(\langle R : FB \rangle)$ absorb $S : R \rightarrow Q$ (in $\text{Alg}(F)$) and becomes $\langle Q \rangle$

When $S : B \rightarrow C$ is an F-homomorphism from $R : F(B) \rightarrow B$ to $Q : F(C) \rightarrow C$, folding with R and then applying S is folding with Q .

(11.4.2a)



§11.5. Type relator

(11.5a)

Let F be a binary relator with initial type (α, T) , so T is a type functor. $F(R, S)$ is its action on a pair, and $F(X)$ abbreviates $F(1, X)$, the F of the reduce section. For every object A the initial algebra is $\alpha : F(A, TA) \rightarrow TA$, among the maps. T acts on an arrow $R : A \rightarrow B$ by

$$T(R) = (F(R, 1) \alpha)$$

(11.5b)

name	law	what it says
the defining equation	$T(R) = (F(R, 1) \alpha)$	Rebuild the structure with α , applying R to the parameter on the way.
functor	$T(1) = 1$ and $T(R)T(S) = T(RS)$	Acting by the identity changes nothing, and two actions in a row are one action.
type functor fusion	$T(f)(h) = (F(f, 1)h)$	A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because F is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$.
naturality of α	$\alpha T(f) = F(f, T(f)) \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R) = F(R, T(R))$ to T .
type relator	$\text{PreservesRecip}(F) \implies T(R)^\circ = T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

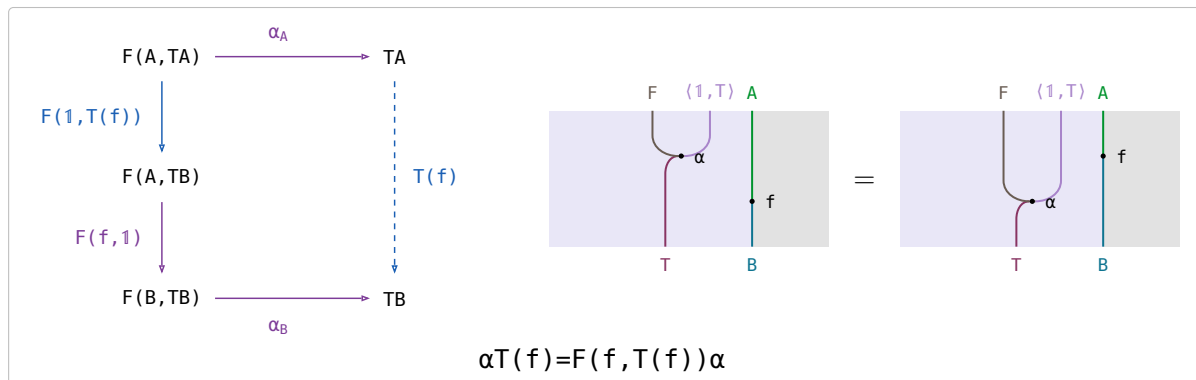
§11.5.1. Type functor

(11.5.1a)

Let F be a bifunctor taking both the parameter A and the recursive position TA , with an initial algebra $\alpha_A : F(A, TA) \rightarrow TA$ for every object A . Then T is a functor, acting on a map $f : A \rightarrow B$ by

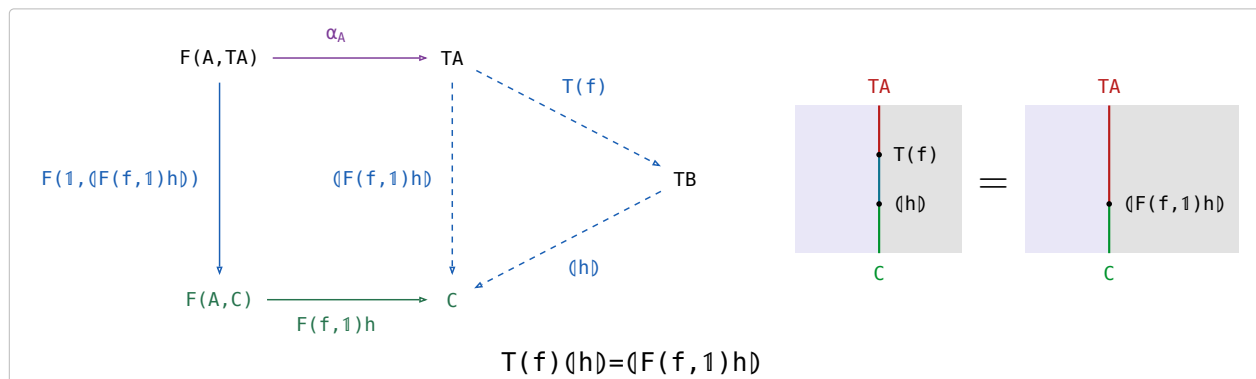
$$T(R) \triangleq (F(R, 1) \alpha)$$

(11.5.1b)



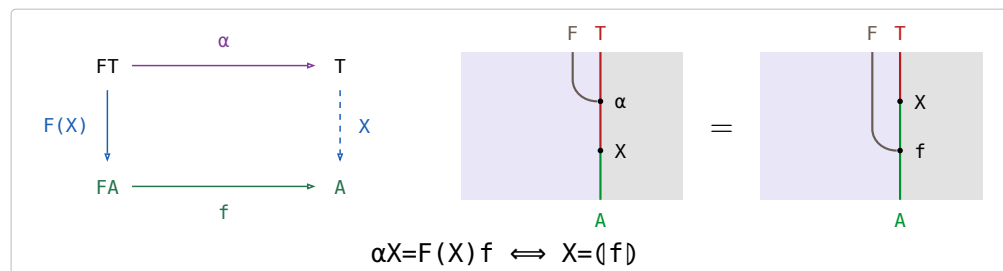
- $F : \mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ is a bifunctor and a wire is a unary functor, so the two arguments are packed first: $\langle 1, T \rangle : \mathcal{A} \rightarrow \mathcal{A} \times \mathcal{A}$ sends A to (A, TA) , and $F(\langle 1, T \rangle(A))$ is $F(A, TA)$.
- The picture is three wires — F , $\langle 1, T \rangle$, and the object — and the region between the first two is $\mathcal{A} \times \mathcal{A}$.
- α is then an ordinary natural transformation $F \circ \langle 1, T \rangle \Rightarrow T$: its bead eats the F and $\langle 1, T \rangle$ wires, and the T wire is born under it.
- $F(f, T(f))$ costs no notation. It is the bead f on the object wire with $\langle 1, T \rangle$ and F running past: $\langle 1, T \rangle$ is what turns f into the pair $(f, T(f))$, and F is what applies it.
- The law is the naturality of α , which is exactly the freedom to slide that f bead past it.

(11.5.1c)



let F be a relator and has **initial algebra** $\alpha : F(T) \rightarrow T$ in the subcategory of functions. α is also initial in the allegory:

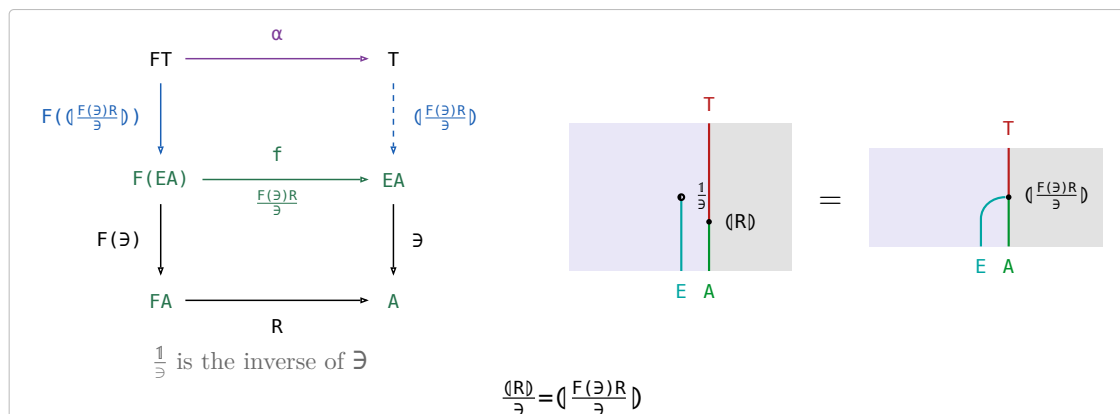
(11.6.1a)



(11.6.1b)

$\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$ catamorphism: arrows of the form $\llbracket f \rrbracket$ are called catamorphisms, and these two examples reveal the notion to be a familiar idea in abstract clothing		
part	Nat	[A]
datatype	$\text{Nat} ::= \text{zero} \mid \text{succ Nat}$	$[A] ::= \text{nil} \mid \text{cons}(A, [A])$
base functor F	$F(X) = 1 + X$	$F(X) = 1 + A \times X$
initial algebra α	$\alpha = [\text{zero}, \text{succ}]$ $: 1 + \text{Nat} \rightarrow \text{Nat}$	$\alpha = [\text{nil}, \text{cons}]$ $: 1 + A \times [A] \rightarrow [A]$
the fold, pointwise	$\llbracket [c, f] \rrbracket(\text{zero}) = c$ $\llbracket [c, f] \rrbracket(\text{succ}(n)) = f(\llbracket [c, f] \rrbracket(n))$	$\llbracket [c, f] \rrbracket(\text{nil}) = c$ $\llbracket [c, f] \rrbracket(\text{cons}(a, x)) = f(a, \llbracket [c, f] \rrbracket(x))$

(11.6.2a)



$$\begin{array}{c}
\boxed{\frac{\alpha X}{F(X)R}} \xleftrightarrow{\cdot \exists \vdash_{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(X)R}{\exists}}} \xleftrightarrow{\cdot \exists \vdash_{\exists}} \boxed{\frac{\frac{\alpha X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}}} \\
\\
\begin{array}{ccccc}
\longleftrightarrow & \boxed{\frac{\alpha \frac{X}{\exists}}{F(\frac{X}{\exists}) \frac{F(\exists)R}{\exists}}} & \longleftrightarrow & \boxed{\frac{\frac{X}{\exists}}{(\frac{F(\exists)R}{\exists})}} & \longleftrightarrow & \boxed{\frac{X}{(\frac{F(\exists)R}{\exists})\exists}} \\
\text{relator, fusion twice} & & \text{reduce of maps} & & &
\end{array}
\end{array}$$

the last two rows at $X := \langle R \rangle$:

$\alpha X = F(X) f \iff X = (f)$ (11.6.2c)		
a relation X out of $\mathbb{T}$ satisfies the defining equation of the fold of f exactly when it is (f)		
formula — each equation of the chain		reason
	$\alpha X = F(X) R$	
	$\frac{\alpha X}{\exists} = \frac{F(X) R}{\exists}$	$\frac{_}{\exists}$ is an isomorphism pow-laws — $\text{frac}(X, \exists) \exists = X$
	$\frac{\alpha X}{\exists} = \frac{F(\frac{X}{\exists}) R}{\exists}$	cancellation, backwards pow-laws — $X = \text{frac}(X, \exists) \exists$
	$\alpha \frac{X}{\exists} = F(\frac{X}{\exists}) \frac{F(\exists) R}{\exists}$	relator; fusion, backwards, twice $\triangleq$ relator — $F(RS) = F(R)F(S)$; pow- laws's fusion at the maps α and $F(\frac{X}{\exists})$
	$\frac{X}{\exists} = (\frac{F(\exists) R}{\exists})$	the fold of a map algebra (11.4a) — the unique map satisfying its defining equation
	$X = (\frac{F(\exists) R}{\exists}) \exists$	cancellation pow-laws — $\text{frac}(X, \exists) \exists = X$

§11.6.3. $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$

(11.6.3a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

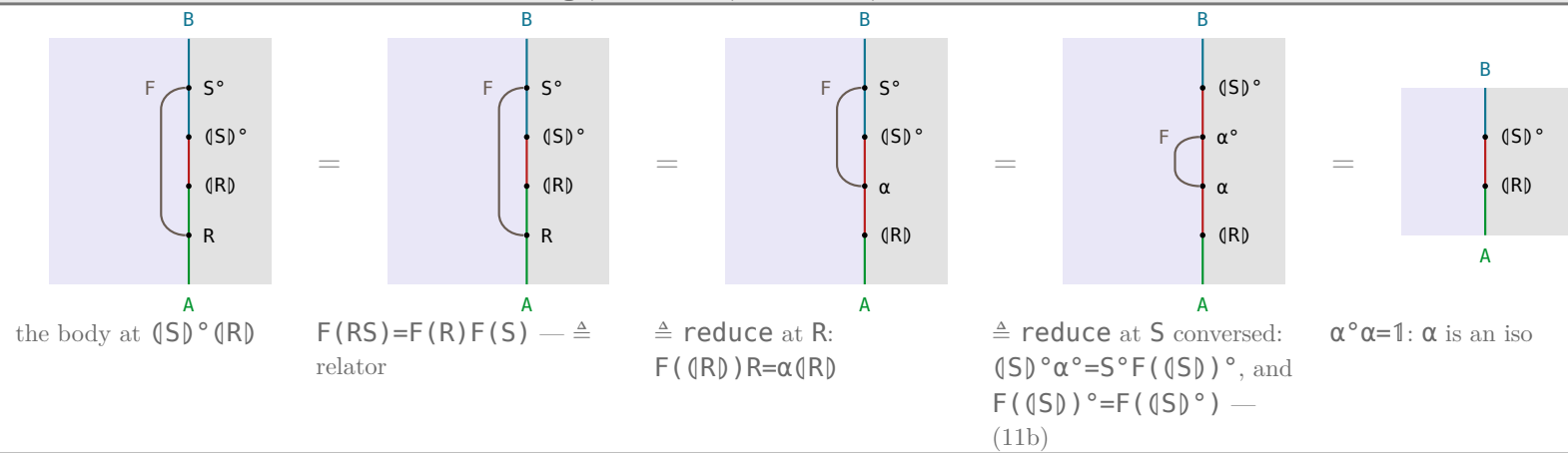
(11.6.3b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 11.6.4 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation

§11.6.4. $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$

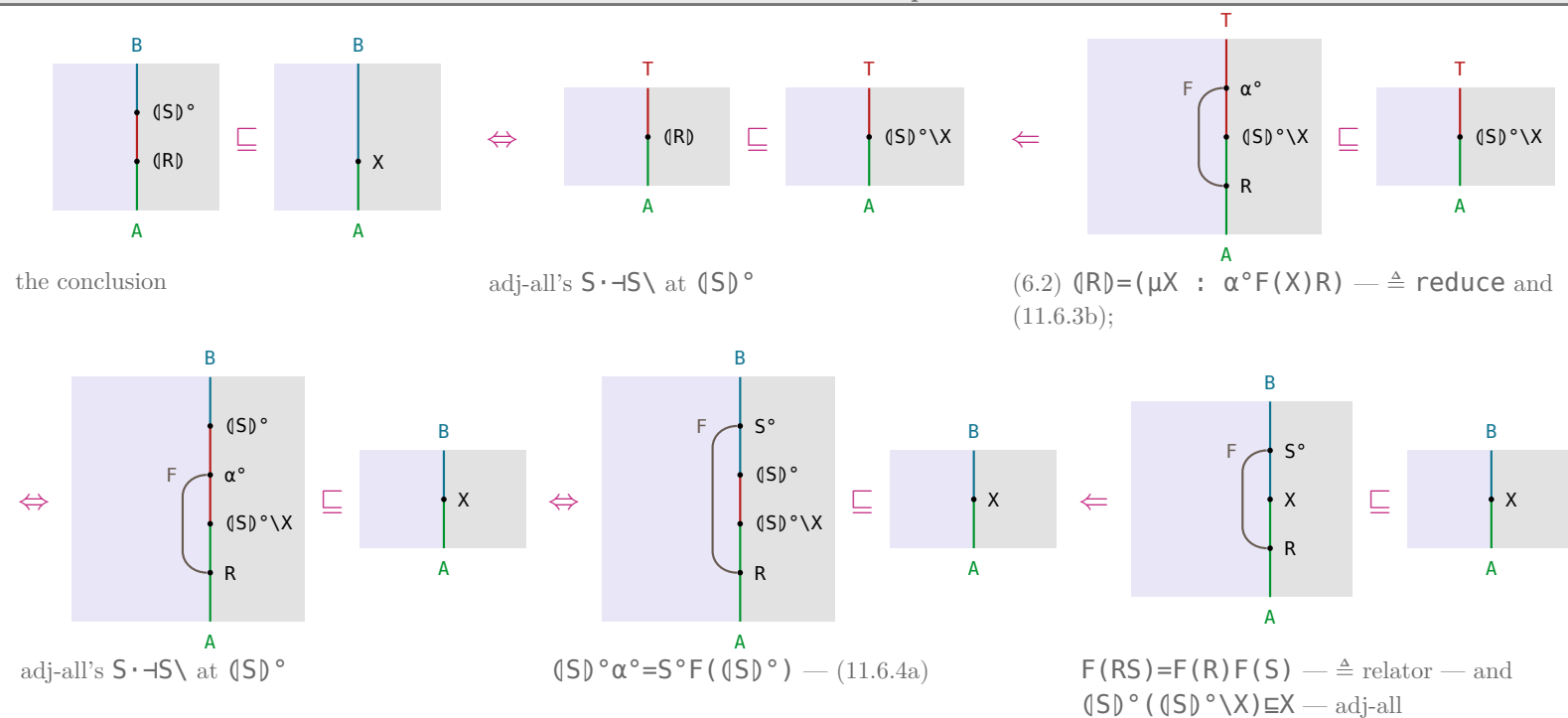
$$\text{PreservesRecip}(F) \Rightarrow S^\circ F((S)^\circ (R)) R = (S)^\circ (R) \quad (11.6.4a)$$

hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((S)^\circ (R))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



$$\text{PreservesRecip}(F) \wedge S^\circ F(X) R \sqsubseteq X \Rightarrow (S)^\circ (R) \sqsubseteq X \quad (11.6.4b)$$

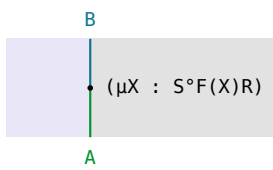
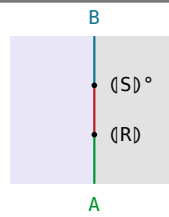
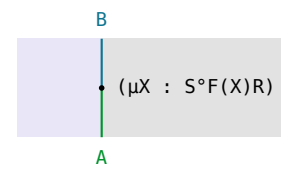
hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(S)^\circ (R)$ is included in X if X satisfies the associated recursion inequation



$$\text{PreservesRecip}(F) \implies \langle S \rangle^\circ \langle R \rangle = (\mu X : S^\circ F(X) R)$$

(11.6.4c)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation


 $\sqsubseteq$

 $\sqsubseteq$

 $\triangleq \mu \text{ at } \varphi(X) := S^\circ F(X) R$

 (11.6.3b)'s $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$

 at $Y := \langle S \rangle^\circ \langle R \rangle$, whose

 $S^\circ F(\langle S \rangle^\circ \langle R \rangle) R = \langle S \rangle^\circ \langle R \rangle$ is (11.6.4a)

 (11.6.4b) at $X := (\mu X : S^\circ F(X) R)$, whose

 $S^\circ F(\mu X : S^\circ F(X) R) R \sqsubseteq (\mu X : S^\circ F(X) R)$ is

 (11.6.3b)'s $\varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$;